Experiment No. 5

Student Name: Ojaswini Thote

PRN: 24070521048

File Path: file:///Users/ojaswinithote/Desktop/Javascript/EXP%205/Practical/index.html
file:///Users/ojaswinithote/Desktop/Javascript/EXP%205/Case%20Study/index1.html

Experiment Title

Apply array methods and object handling; create a cart total calculator with discount logic.

Software / Tools Required

1. Visual Studio Code
2. Google Chrome
3. HTML5
4. JavaScript (ES6)

Task 5.1 — Shopping Cart Calculator

Aim

Demonstrate JavaScript array methods — forEach, map, filter, and reduce — using a Shopping Cart Calculator with product objects.

File Path

file:///Users/ojaswinithote/Desktop/Javascript/EXP%205/Practical/index.html

Program Code

Task5/5.1/index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Shopping Cart Total Calculator</title>

<style>

*{
    margin:0;
    padding:0;
    box-sizing:border-box;
    font-family:Arial, Helvetica, sans-serif;
}

body{
    background:#ffe6f2;
    padding:30px;
}
```

```css
.container{
    width:95%;
    max-width:1400px;
    margin:auto;
    background:white;
    border-radius:15px;
    box-shadow:0 5px 15px rgba(0,0,0,0.2);
    padding:30px;
}

h1{
    text-align:center;
    color:#d63384;
    margin-bottom:30px;
}

.main{
    display:flex;
    gap:30px;
    align-items:flex-start;
}

.left{
    flex:1;
}

.right{
    flex:1;
}

label{
    display:block;
    margin-top:15px;
    margin-bottom:5px;
    font-weight:bold;
}

input{
    width:100%;
    padding:10px;
    border:1px solid #ccc;
    border-radius:8px;
    font-size:16px;
}

button{
    width:100%;
    padding:12px;
    margin-top:15px;
    background:#ff4d94;
    color:white;
    border:none;
    border-radius:8px;
    cursor:pointer;
    font-size:16px;
}

button:hover{
```

```
    background:#e60073;
}

.right h2{
    color:#d63384;
    margin-bottom:15px;
}

.product{
    background:#fff0f6;
    border-left:5px solid #ff4d94;
    padding:12px;
    margin-bottom:12px;
    border-radius:8px;
    line-height:1.7;
}

#result{
    margin-top:30px;
    background:#ffeaf3;
    padding:20px;
    border-radius:10px;
    font-size:18px;
    line-height:1.8;
    font-weight:bold;
}

hr{
    margin:12px 0;
}

@media(max-width:900px){

.main{
    flex-direction:column;
}

}

</style>

</head>
<body>

<div class="container">

<h1>□ Shopping Cart Total Calculator</h1>

<div class="main">

<div class="left">

<label>Product Name</label>
<input type="text" id="productName" placeholder="Enter Product Name">

<label>Price (₹)</label>
<input type="number" id="price" placeholder="Enter Price">
```

```html
<label>Quantity</label>
<input type="number" id="quantity" placeholder="Enter Quantity">

<button onclick="addProduct()">Add Product</button>

<button onclick="calculateBill()">Calculate Total</button>

</div>

<div class="right">

<h2>Products Added</h2>

<div id="productList">
<p>No products added.</p>
</div>

</div>

</div>

<div id="result"></div>

</div>

<script src="script.js"></script>

</body>
</html>
```

Task5/5.1/script.js

```javascript
// Array to store product objects
const cart = [];

// Add Product
function addProduct(){

    const name = document.getElementById("productName").value.trim();
    const price = Number(document.getElementById("price").value);
    const quantity = Number(document.getElementById("quantity").value);

    if(name==="" || price<=0 || quantity<=0){
        alert("Please enter valid product details.");
        return;
    }

    // Object
    const product={
        name:name,
        price:price,
        quantity:quantity,
        total:price*quantity
    };

    // Array Method
```

```
    cart.push(product);

    displayProducts();

    document.getElementById("productName").value="";
    document.getElementById("price").value="";
    document.getElementById("quantity").value="";
}

// Display Products
function displayProducts(){

    const productList=document.getElementById("productList");

    productList.innerHTML="";

    cart.forEach(function(item,index){

        productList.innerHTML+=`
        <div class="product">
            <strong>${index+1}. ${item.name}</strong><br>
            Price : ₹${item.price}<br>
            Quantity : ${item.quantity}<br>
            Total : ₹${item.total}
        </div>
        `;

    });

}

// Calculate Bill
function calculateBill(){

    if(cart.length===0){
        alert("Cart is empty.");
        return;
    }

    // Array Method - reduce()
    const cartTotal=cart.reduce(function(sum,item){
        return sum+item.total;
    },0);

    let discount=0;

    if(cartTotal>=5000){
        discount=cartTotal*0.20;
    }
    else if(cartTotal>=3000){
        discount=cartTotal*0.15;
    }
    else if(cartTotal>=1000){
        discount=cartTotal*0.10;
    }

    const finalAmount=cartTotal-discount;
```

```
        document.getElementById("result").innerHTML=`
            <h2 style="color:#d63384;">Bill Summary</h2>
            Total Products : ${cart.length}<br>
            Cart Total : ₹${cartTotal.toFixed(2)}<br>
            Discount : ₹${discount.toFixed(2)}
            <hr>
            Final Amount : ₹${finalAmount.toFixed(2)}
        `;

    }
```

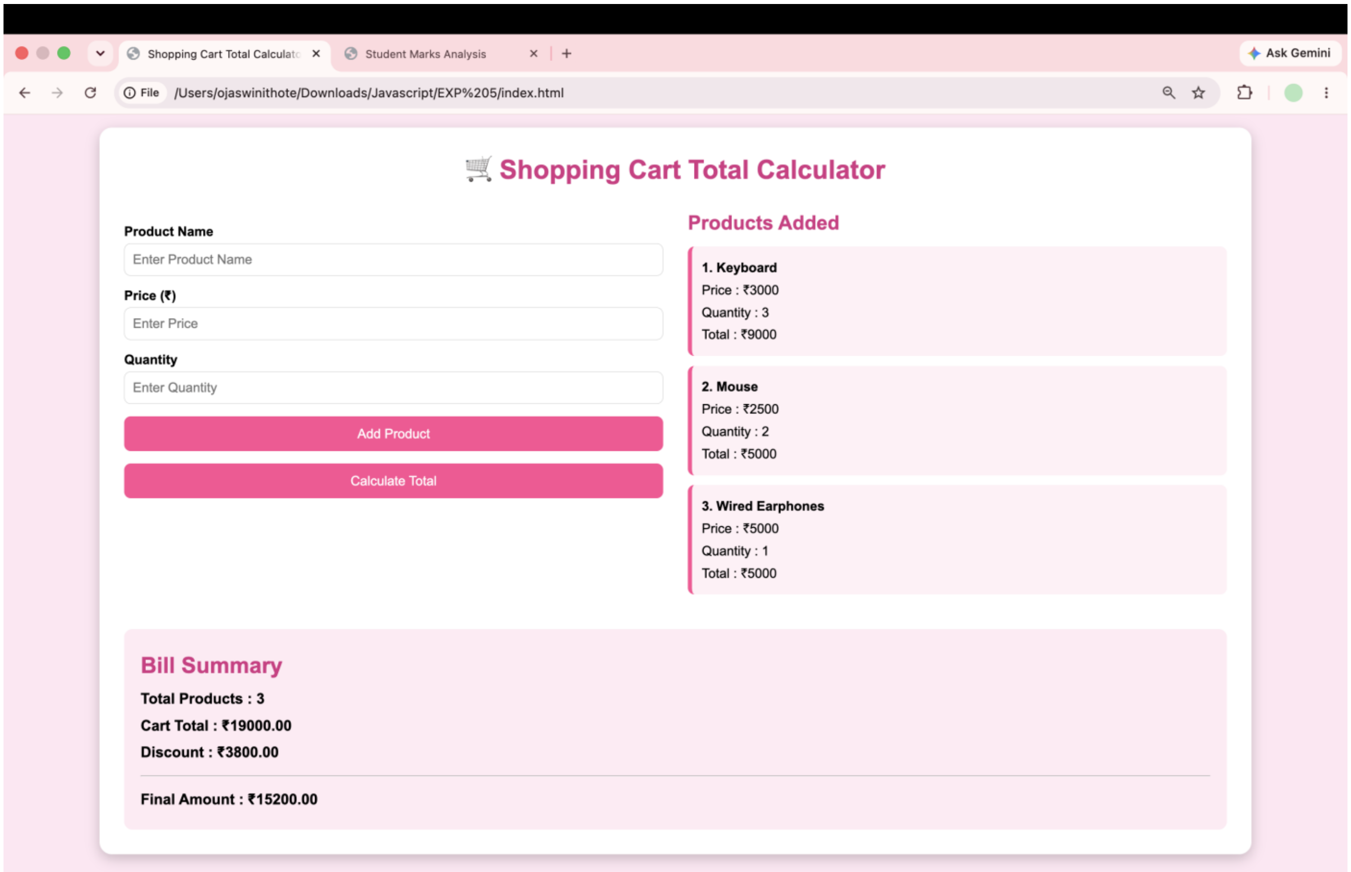
Array Methods Used

Method	**Purpose**
forEach()	Renders each cart item as a table row
map()	Builds the Item Summary list with product name and item total
filter()	Extracts products where price > 1000
reduce()	Calculates the grand total: sum + (price × quantity)

Output

- User enters **Product Name, Price, and Quantity** and clicks **Add Product**.
- The table displays **ID, Name, Price, Quantity, and Total** for each product.
- Discount is applied automatically:
- **≥ ₹50,000 → 20% discount**
- **≥ ₹20,000 → 10% discount**
- **≥ ₹5,000 → 5% discount**
- **Final Amount = Total − Discount**
- Item Summary and Expensive Products lists are updated whenever a new product is added.

Screenshot



Task 5.2 — Max & Min Finder (Array of Objects)

Aim

Create an array of objects from user input and find the Maximum and Minimum values using array methods — map(), Math.max(), Math.min(), and some().

Program Code

Task5/5.2/index.html

```
<!DOCTYPE html>
<html lang="en">

<head>

<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Student Marks Analysis</title>

<style>

*{
    margin:0;
    padding:0;
    box-sizing:border-box;
    font-family:'Segoe UI',Tahoma,Geneva,Verdana,sans-serif;
```

```css
}

body{
    min-height:100vh;
    padding:40px;
    background:linear-gradient(135deg,#4facfe,#00f2fe);
}

.container{
    width:95%;
    max-width:1100px;
    margin:auto;
    background:white;
    padding:40px;
    border-radius:20px;
    box-shadow:0 15px 40px rgba(0,0,0,.25);
}

h1{
    text-align:center;
    color:#0d47a1;
    margin-bottom:10px;
    font-size:34px;
}

.subtitle{
    text-align:center;
    color:#666;
    margin-bottom:35px;
}

.form-section{
    background:#f4f8ff;
    padding:25px;
    border-radius:15px;
    border-left:6px solid #1565c0;
}

.form-section h2{
    color:#0d47a1;
    margin-bottom:20px;
}

.input-row{
    display:flex;
    gap:20px;
}

.input-group{
    flex:1;
}

label{
    display:block;
    font-size:16px;
    font-weight:600;
    margin-bottom:8px;
    color:#333;
}

input{
    width:100%;
    padding:13px;
    border:2px solid #90caf9;
    border-radius:10px;
    font-size:16px;
    outline:none;
```

```
}

input:focus{
    border-color:#1565c0;
    box-shadow:0 0 8px rgba(21,101,192,.25);
}

button{
    padding:13px;
    border:none;
    border-radius:10px;
    background:linear-gradient(135deg,#1976d2,#0d47a1);
    color:white;
    font-size:16px;
    font-weight:bold;
    cursor:pointer;
    transition:.3s;
}

button:hover{
    transform:translateY(-2px);
    box-shadow:0 7px 15px rgba(13,71,161,.3);
}

.add-button{
    width:100%;
    margin-top:20px;
}

.operations{
    display:grid;
    grid-template-columns:1fr 1fr;
    gap:15px;
    margin-top:20px;
}

.operations button{
    width:100%;
}

.analyze{
    width:100%;
    margin-top:20px;
    background:linear-gradient(135deg,#43a047,#1b5e20);
}

.students-count{
    text-align:center;
    margin-top:20px;
    color:#555;
    font-size:16px;
}

.results{
    display:flex;
    gap:25px;
    margin-top:35px;
}

.card{
    flex:1;
    padding:30px;
    border-radius:15px;
    text-align:center;
    box-shadow:0 8px 20px rgba(0,0,0,.12);
}
```

```css
.topper{
    background:linear-gradient(135deg,#fff8dc,#ffe082);
}

.lowest{
    background:linear-gradient(135deg,#ffebee,#ef9a9a);
}

.icon{
    font-size:50px;
    margin-bottom:10px;
}

.card h2{
    margin-bottom:15px;
}

.card h3{
    font-size:25px;
    margin-bottom:8px;
}

.card p{
    font-size:17px;
    margin:5px 0;
    color:#444;
}

.marks{
    font-size:32px !important;
    font-weight:bold;
    margin-top:15px !important;
}

#message{
    text-align:center;
    margin-top:20px;
    font-weight:bold;
    color:#1565c0;
}

@media(max-width:700px){

    body{
        padding:20px;
    }

    .container{
        width:100%;
        padding:25px;
    }

    .input-row{
        flex-direction:column;
    }

    .operations{
        grid-template-columns:1fr;
    }

    .results{
        flex-direction:column;
    }

}
```

```html
</style>

</head>


<body>

<div class="container">

    <h1>Student Marks Analysis</h1>

    <p class="subtitle">
        Student Performance using Arrays, Objects and Array Methods
    </p>


    <div class="form-section">

        <h2>Enter Student Details</h2>


        <div class="input-row">

            <div class="input-group">

                <label for="name">
                    Student Name
                </label>

                <input
                    type="text"
                    id="name"
                    placeholder="Enter student name"
                >

            </div>


            <div class="input-group">

                <label for="prn">
                    PRN
                </label>

                <input
                    type="text"
                    id="prn"
                    placeholder="Enter PRN"
                >

            </div>


            <div class="input-group">

                <label for="marks">
                    Marks
                </label>

                <input
                    type="number"
                    id="marks"
                    placeholder="Enter marks"
                    min="0"
```

```html
                max="100"
            >

        </div>

    </div>


    <!-- PUSH -->

    <button class="add-button" onclick="addStudent()">
        Add Student (push)
    </button>


    <!-- POP, UNSHIFT, SHIFT -->

    <div class="operations">

        <button onclick="removeLastStudent()">
            Remove Last Student (pop)
        </button>

        <button onclick="removeFirstStudent()">
            Remove First Student (shift)
        </button>

        <button onclick="addFirstStudent()">
            Add Student at Beginning (unshift)
        </button>

        <button onclick="showCurrentArray()">
            Check Array in Console
        </button>

    </div>


    <button class="analyze" onclick="analyzeStudents()">
        Analyze Student Performance
    </button>


    <div class="students-count" id="studentCount">
        No students added yet.
    </div>

</div>


<!-- FINAL RESULTS ONLY -->

<div class="results">

    <div class="card topper">

        <div class="icon">□</div>

        <h2>Topper</h2>

        <h3 id="topperName">---</h3>

        <p id="topperPRN">
            PRN: ---
```

```
            </p>

            <p class="marks" id="topperMarks">
                --- Marks
            </p>

        </div>


        <div class="card lowest">

            <div class="icon">□</div>

            <h2>Lowest Marks</h2>

            <h3 id="lowestName">---</h3>

            <p id="lowestPRN">
                PRN: ---
            </p>

            <p class="marks" id="lowestMarks">
                --- Marks
            </p>

        </div>

    </div>


    <div id="message"></div>

</div>


<script src="script1.js"></script>

</body>

</html>
```

Task5/5.2/script.js

```
let students = [];


// ==========================================
// PUSH - ADD STUDENT
// ==========================================

function addStudent(){

    let name = document.getElementById("name").value.trim();
    let prn = document.getElementById("prn").value.trim();
    let marks = Number(document.getElementById("marks").value);

    let message = document.getElementById("message");


    if(name === "" || prn === "" || isNaN(marks)){

        message.textContent =
            "Please enter all student details.";
```

```
        return;
    }


    if(marks < 0 || marks > 100){

        message.textContent =
            "Marks must be between 0 and 100.";

        return;
    }


    // Create Student Object

    let student = {

        name: name,
        prn: prn,
        marks: marks

    };


    // PUSH
    students.push(student);


    console.log("PUSH() executed");
    console.log("Student added:", student);
    console.log("Current Array:", students);


    updateStudentCount();


    document.getElementById("name").value = "";
    document.getElementById("prn").value = "";
    document.getElementById("marks").value = "";

    message.textContent =
        "Student added successfully using push().";

}


// ==========================================
// POP - REMOVE LAST STUDENT
// ==========================================

function removeLastStudent(){

    if(students.length === 0){

        document.getElementById("message").textContent =
            "No students available.";

        return;
    }


    let removedStudent = students.pop();
```

```
    console.log("POP() executed");
    console.log("Removed Student:", removedStudent);
    console.log("Current Array:", students);


    updateStudentCount();


    document.getElementById("message").textContent =
        removedStudent.name +
        " removed using pop().";

}


// ==========================================
// UNSHIFT - ADD STUDENT AT BEGINNING
// ==========================================

function addFirstStudent(){

    let name = document.getElementById("name").value.trim();
    let prn = document.getElementById("prn").value.trim();
    let marks = Number(document.getElementById("marks").value);


    if(name === "" || prn === "" || isNaN(marks)){

        document.getElementById("message").textContent =
            "Enter student details first.";

        return;
    }


    if(marks < 0 || marks > 100){

        document.getElementById("message").textContent =
            "Marks must be between 0 and 100.";

        return;
    }


    let student = {

        name: name,
        prn: prn,
        marks: marks

    };


    // UNSHIFT

    students.unshift(student);


    console.log("UNSHIFT() executed");
    console.log("Student added at beginning:", student);
    console.log("Current Array:", students);
```

```
    updateStudentCount();


    document.getElementById("name").value = "";
    document.getElementById("prn").value = "";
    document.getElementById("marks").value = "";


    document.getElementById("message").textContent =
        "Student added at beginning using unshift().";

}


// ==========================================
// SHIFT - REMOVE FIRST STUDENT
// ==========================================

function removeFirstStudent(){

    if(students.length === 0){

        document.getElementById("message").textContent =
            "No students available.";

        return;
    }


    let removedStudent = students.shift();


    console.log("SHIFT() executed");
    console.log("Removed Student:", removedStudent);
    console.log("Current Array:", students);


    updateStudentCount();


    document.getElementById("message").textContent =
        removedStudent.name +
        " removed using shift().";

}


// ==========================================
// FOREACH
// ==========================================

function analyzeStudents(){

    if(students.length === 0){

        document.getElementById("message").textContent =
            "Please add students first.";

        return;
    }
```

```
console.log("================================");
console.log("ARRAY METHODS");
console.log("================================");


// ======================================
// forEach()
// ======================================

console.log("forEach() executed:");

students.forEach(function(student){

    console.log(
        student.name +
        " | " +
        student.prn +
        " | " +
        student.marks
    );

});


// ======================================
// map()
// ======================================

let graceMarks = students.map(function(student){

    return {

        name: student.name,
        prn: student.prn,
        marks: student.marks + 5

    };

});


console.log("map() executed:");
console.log("Grace Marks:", graceMarks);


// ======================================
// filter()
// ======================================

let passedStudents = students.filter(function(student){

    return student.marks >= 40;

});


console.log("filter() executed:");
console.log("Passed Students:", passedStudents);


// ======================================
// reduce()
// ======================================
```

```
let totalMarks = students.reduce(function(total, student){

    return total + student.marks;

}, 0);


console.log("reduce() executed:");
console.log("Total Marks:", totalMarks);


// =====================================
// AVERAGE
// =====================================

let average = totalMarks / students.length;


console.log("Average Marks:", average);


// =====================================
// FIND HIGHEST MARKS
// =====================================

let highestMarks = Math.max(

    ...students.map(function(student){

        return student.marks;

    })

);


// =====================================
// FIND LOWEST MARKS
// =====================================

let lowestMarks = Math.min(

    ...students.map(function(student){

        return student.marks;

    })

);


// =====================================
// FIND TOPPER
// =====================================

let topper = students.find(function(student){

    return student.marks === highestMarks;

});


// =====================================
```

```
    // FIND LOWEST STUDENT
    // ======================================

    let lowestStudent = students.find(function(student){

        return student.marks === lowestMarks;

    });


    console.log("Topper:", topper);

    console.log("Lowest Marks Student:", lowestStudent);


    // ======================================
    // DISPLAY FINAL RESULTS
    // ======================================

    document.getElementById("topperName").textContent =
        topper.name;

    document.getElementById("topperPRN").textContent =
        "PRN: " + topper.prn;

    document.getElementById("topperMarks").textContent =
        topper.marks + " Marks";


    document.getElementById("lowestName").textContent =
        lowestStudent.name;

    document.getElementById("lowestPRN").textContent =
        "PRN: " + lowestStudent.prn;

    document.getElementById("lowestMarks").textContent =
        lowestStudent.marks + " Marks";


    document.getElementById("message").textContent =
        "Analysis completed successfully.";

}


// ==========================================
// DISPLAY ARRAY IN CONSOLE ONLY
// ==========================================

function showCurrentArray(){

    console.log("Current Student Array:");
    console.table(students);

    document.getElementById("message").textContent =
        "Current array displayed in browser console.";

}


// ==========================================
// UPDATE STUDENT COUNT
// ==========================================
```

```
function updateStudentCount(){

    document.getElementById("studentCount").textContent =
        students.length +
        " student(s) currently stored.";

}
```

Array Methods Used

Method	Purpose
map()	Converts input strings into an array of number objects
map()	Extracts value from each object into a plain array
some()	Validates that no entry is NaN
Math.max()	Finds the largest number using the spread operator
Math.min()	Finds the smallest number using the spread operator

Output

- User enters comma-separated numbers.

Example:

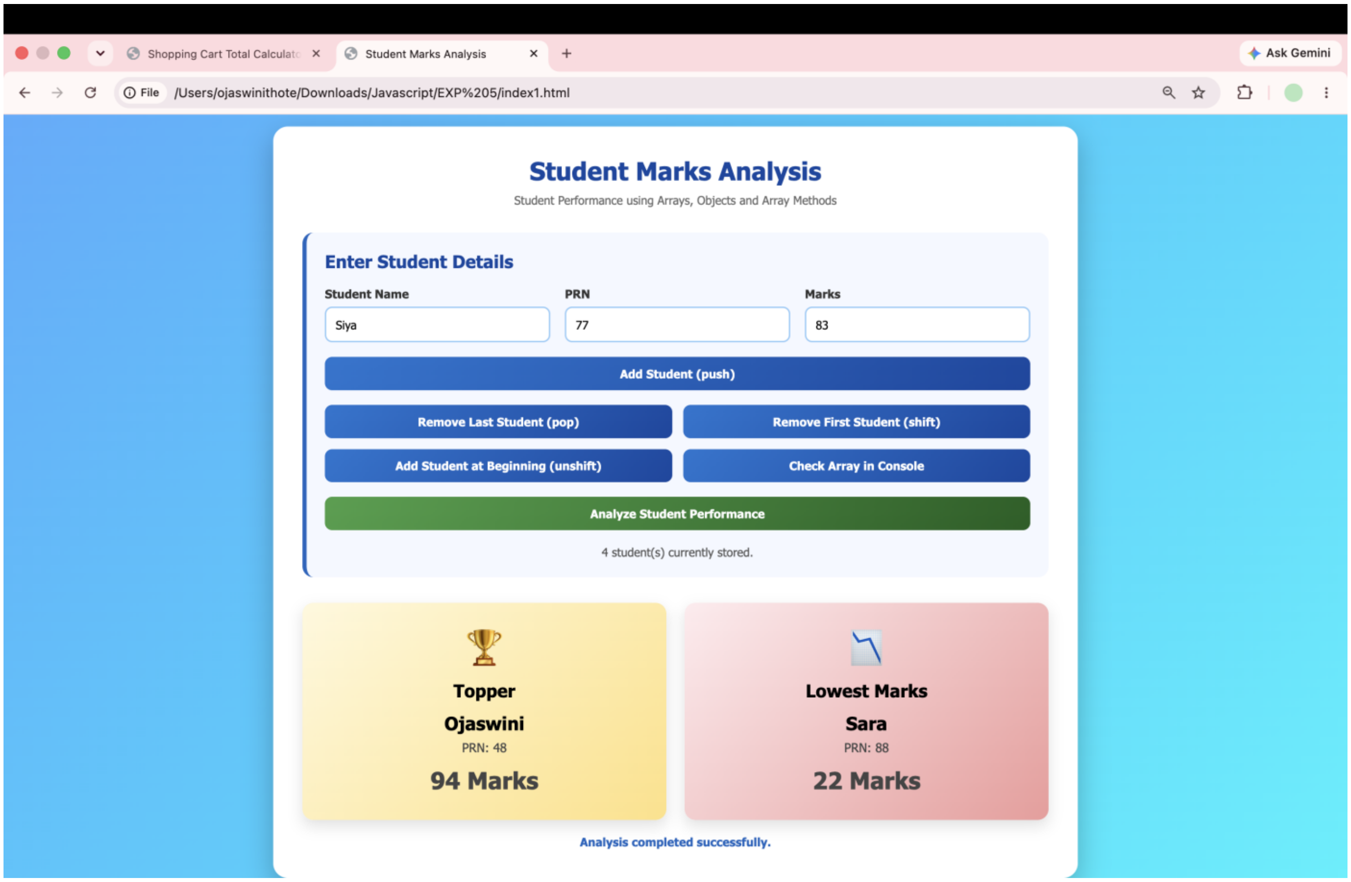
```
10, 25, 3, 47, 8
```

- Clicking **Find Max & Min** displays:

```
Numbers Array: [10, 25, 3, 47, 8]
Maximum Value: 47
Minimum Value: 3
```

- Invalid input displays an error message in red.

Screenshot



Result / Conclusion

Both tasks were completed successfully.

- **Task 5.1** demonstrated forEach(), map(), filter(), and reduce() through a Shopping Cart Calculator that dynamically adds products, computes totals, applies discounts, and lists expensive products.
- **Task 5.2** demonstrated map(), some(), Math.max(), and Math.min() by converting user-entered comma-separated numbers into an array of objects and finding the maximum and minimum values.

Thus, the experiment successfully demonstrated the practical use of **JavaScript Array Methods and Object Handling** in web applications.